



THE COPPERBELT UNIVERSITY

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TITLE:

EVALUATING ONLINE HEALTH-SEEKING BEHAVIORS AS DIGITAL PROXIES
FOR EMERGING INFECTIOUS DISEASE BURDEN IN ZAMBIA: A GOOGLE TRENDS
TIME-SERIES ANALYSIS (2019–2024)

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THE DEPARTMENT OF PUBLIC HEALTH

THIS RESEARCH PROPOSAL IS BEING SUBMITTED IN PARTIAL FULFILMENT OF
THE REQUIREMENT FOR THE BACHELOR OF SCIENCE IN PUBLIC HEALTH

DECLARATION

I Taonga Jere student number 21166977 and NRC number, do at this moment, declare that this research proposal entitled, **evaluating online health-seeking behaviors as digital proxies for emerging infectious disease burden in zambia: a google trends time-series analysis (2019–2024)**, which is being submitted to the Public Health Department of the Copperbelt University School, of Medicine, in partial fulfillment of the requirements for the award of a degree in the bachelor of science in public health, is my authentic work and has not been presented to any university or institution for award of any degree.

Signature:

Date:

CERTIFICATION FOR APPROVAL

STUDENT

I Taonga Jere hereby state that this research proposal is the result of my effort. The various sources, to which I will forever be indebted, have been indicated in the citations and references.

Signed:

Date:

SUPERVISOR

I, hereby state that I have checked this project proposal is adequate in terms of scope and quality. I, therefore, approve that it should be submitted for assessment.

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Date:

ABSTRACT

Zambia has continued to experience a huge burden of emerging infectious. An emerging infectious disease is one that either has appeared and affected a population for the first time, or has existed previously but is rapidly spreading, either in terms of the number of people getting infected, or to new geographical areas. Traditional surveillance methods, while necessary, are often hindered by delayed reporting and lack real-time data on population health behaviors. As internet use for health information grows, digital data sources like Google search trends (GST) offer a useful supplement for public health monitoring.

This study aims to examine the relationship between Google Relative Search Volume (RSV) and emerging infectious diseases (EID) in Zambia from 2019 to 2024. Specifically, it will analyze trends in EID-related search queries and compare them with reported EID data from the Health Management Information System (HMIS)/District Health Information Software 2 (DHIS2). A quantitative ecological time-series design will be used, including descriptive trend analysis, Pearson correlation, lagged correlation, and regression analysis to assess the strength and direction of the relationship.

The study also examines contextual factors such as internet access, population size, and major health events that may affect search behavior. By evaluating the potential of using Google Trends as an additional public health surveillance tool, this research aims to contribute to the expanding field of digital epidemiology and offer insights that could support more timely and data-driven health responses in Zambia.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BREC	Biomedical Research Ethics Committee
CBU	Copperbelt University
CDC	Centers for Disease Control and Prevention
CSO	Central Statistical Office
DHIS2	District Health Information System 2
DRC	Democratic Republic of Congo
EID	Emerging Infectious Disease
GDPR	General Data Protection Regulation
GHT	Google Health Trends
GST	Google Search Trends
HMIS	Health Management Information System
MOH	Ministry of Health
RSV	Relative Search Volume
SDG	Sustainable Development Goals
TPB	Theory of Planned Behavior
WHO	World Health Organization
ZICTA	Zambia Information and Communications Technology Authority

CHAPTER ONE INTRODUCTION

1.1 BACKGROUND

Zambia has continued to experience a huge burden of disease, mainly characterized by: high prevalence and impact of communicable diseases, particularly malaria, HIV and AIDS, Sexually Transmitted Infections (STIs) and Tuberculosis (TB); and high maternal, neonatal and child morbidity and mortality (Ministry of Health, 2023).

EID account for a majority of deaths in Zambia; As of 2021, 52% of deaths were due to infectious diseases (World Health Organization (WHO), 2024). For years, population-based surveys and clinical data have been the primary sources of disease risk factor surveillance, and these traditional methods have surely played an important role in improving early detection; However, these methods are often restricted by challenges such as low response rates, recall bias, and long data collection cycles (Li *et al.*, 2022). Thus, there is a need to explore innovative data sources and collection techniques to strengthen disease risk factor surveillance (Bulage *et al.*, 2022).

While EIDs are serious public health threats (World Health Organization (WHO), 2016), and traditional surveillance methods lack adequate monitoring of their prevalence and impact, innovative techniques, such as Google Trends Analytics, have emerged as promising complementary tools for improving public health surveillance (Tobias Seagner and Donatas Austys, 2020).

By analysing search query patterns, Google Trends Analytics is a potential approach for monitoring public health interests, awareness, and behaviours related to disease risk factors (Tobias Seagner and Donatas Austys, 2020). While in its infancy in disease risk factor monitoring, several studies have explored its feasibility by exploring the use of search trends to explore public interests and correlate it with incidence. For example, a study was conducted and its primary focus was to explore the correlation between GHT search query data and reported COVID-19 case incidence across 54 African countries, and found that GHT search query data showed a weak correlation with reported COVID-19 case incidence across most African countries, suggesting limited overall applicability of GHT as a surveillance tool with COVID-19; due to, in part, of the unpredictable and novel nature of the disease, and noting that GHT may yield stronger results for diseases with consistent, seasonal patterns of incidence (Alexander Fulk, 2022).

1.2 PROBLEM STATEMENT

EIDs are diseases that either have appeared and affected a population for the first time, or have existed previously but are rapidly spreading, either in terms of the number of people getting infected, or to new geographical areas (World Health Organization (WHO), 2016). Zambia experiences a huge burden of EIDs (Ministry of Health, 2023). While global health systems are rapidly increasing the adoption of data science methods and technologies like Artificial Intelligence (AI), Zambia is lagging behind in adopting these technologies, despite facing a growing burden of disease (Ministry of Health (MOH), 2017). At the same time, many Zambians are using the internet to search for health-related information, including symptoms, treatments, and prevention (Malauzi Felesia *et al.*, 2020).

However, there is a need for reform in digital health governance, sustainable investment, workforce development, enhanced policy frameworks, and infrastructure improvements in Zambia (Transform Health, 2025). As a result, without robust digital health systems, the availability and timeliness of health data are limited, thereby constraining effective surveillance and decision-making (The Global Fund and Jason Mulikita, 2025).

This study aims to assess the correlation between GSV data and public health interest in EIDs. A successful strong and consistent correlation, could demonstrate the benefits of integrating online search trends into public health surveillance systems, helping health authorities plan awareness campaigns more effectively and align them with real-time public concerns.

1.3 RATIONALE FOR THE STUDY

This study is based on the increasing demand for digital data in public health. More people are turning to digital sources for health information, which makes GST a reliable source of insights into public health information and data (Pawar *et al.*, 2020).

Using GST as a tool for public health surveillance can bridge the gap between online search behavior and formal health care (Asch *et al.*, 2019). By allowing researchers to identify when interest in EIDs peaks and when it declines, guiding targeted education and prevention campaigns.

Furthermore, this kind of study can pave the way for demonstrating the feasibility of using big data tools, such as social media analytics, machine learning, and predictive models, alongside traditional surveillance methods in Zambia. If proven effective, this approach could be scaled up to cover all health conditions, fostering a more digital and data-driven public health system.

1.4 LITERATURE REVIEW

1.4.1 GLOBAL PERSPECTIVE

Globally, access to the internet has transformed how people seek medical advice -they Google their symptoms, diagnoses, and treatment options, because they believe it is reliable, fast, and cost-effective (Aoun, Lakkis and Antoun, 2020). This leaves traces -search queries, and data is generated, which reflects their digital phenotype (Dlima *et al.*, 2022). This data can be used to better understand people's online behaviour, and further analysis can predict what people are concerned about in a given moment in time. Researchers began to investigate whether these trends can be used as public health indicators, which can be used in integration or completely innovate traditional surveillance methods, which are unreliable and, in most cases, slow (Mremi *et al.*, 2021).

Early digital epidemiology paved the way for the use of search query data as a public health indicator. Neumann *et al.* (2022) investigated the use of GHT – API for epidemiological research, arguing that search data is timely and accessible and offers potential for this field. The researchers explored how internet health data can reflect exposures or health outcomes by evaluating whether Google Search data can be used beyond simple monitoring tools. Another research by Shepherd, Robinson and Mallen, (2023) found that there were large spikes in search volume data following the announcement of MPOX cases, which shows how search behaviour reflects public awareness and concern.

GHT data, as a public health indicator, has broadened its scope from human infectious diseases to zoonotic diseases. Wisnieski *et al.*, (2023) argued that Google Trends holds promise for zoonotic disease prediction, but is underutilised and needs methodological refinement. Similarly, a study by Dong-Her Shih *et al.*, (2024) on infodemiology of influenza-like illnesses incorporated GSV alongside climatic variables, which improved prediction modelling for routine acute diseases and highlighted the adaptability of infovellance in multiple epidemiologic contexts.

A study conducted by Raubenheimer, Myburgh and Bhagavathula, (2024) analysed worldwide search interest in semaglutide -a drug used for diabetes and weight loss, from January 2021 to August 2023. They assessed how search behaviour reflected interests in emerging treatments for diseases and how social influence might have driven search volumes. Collectively, these studies showcase how progression in the integration of GHT

data in health surveillance has evolved, from its use in validating search query data as a reliable epidemiologic tool, to applying it for emerging and routine infectious diseases, and finally, extending its use to non-communicable diseases and public engagement in health interventions. This evolution stresses the relevance of GHT use as a public health indicator as it offers data that is timely, cost-effective, and that shows population interests in real time.

1.4.2 SOUTHERN AFRICAN PERSPECTIVE

A study by Irene R Mremi *et al.*, (2021) evaluated existing surveillance systems in the region and revealed persistent challenges that underscored the value of digital approaches. Early research in sub-Saharan Africa highlights the opportunity offered by big data analytics to enhance infectious disease monitoring, and increasing its recognition of its potential to improve public health surveillance, mitigating these surveillance system challenges. Achieng and Ogundaini, (2024), for example, conducted a systematic review of 10 studies via PRISMA methodology and identified how analytics could improve timeliness and integration of surveillance given frequent outbreaks, which emphasised the potential of big data analytics for infectious disease surveillance.

Building on this foundation, researchers begun exploring practical applications of digital data to monitor population health. A study by Oladeji *et al.*, (2021) explored the use of Google search data to monitor diet, exercise, and obesity in Africa, by analysing Google search terms for 52 countries, and compared them with WHO prevalence data. By finding a strong correlation between the two data sets, they demonstrated that internet search behaviour could act as a proxy for population-level risk factors. Similarly, Alexander Fulk, *et al.*, (2022) investigated whether search data can anticipate outbreaks by collecting search volume data for COVID-19 and coronavirus, and compared them with weekly incidence. Although they concluded that effectiveness depended on internet access and case reporting quality, this showed that GHT data could be used to track COVID-19 information-seeking patterns.

Although the integration of GHT into public health surveillance has proved to be a promising field. This approach heavily relies on the reliability of digital infrastructure and disease burden. A study conducted by Tsenahun Manyazewai *et al.* (2023), which assessed African countries' digital infrastructure, with the rationale that investments in digital health

technologies could help Africa meet the UN's SDG 3 health targets by 2030, but such deployment is dependent on the digital infrastructure and the disease burden context in each country. Such research strengthens the idea that the success of digital health technologies doesn't only depend on its introduction, but also on its underlying ecosystem; internet access, digital literacy, investment, and policy. This, however, does not mean digital health care should be a venture only for countries with robust digital literacy or access. Africa CDC, (2023) digital disease surveillance programme emphasises that traditional surveillance alone is insufficient, campaigning for a more digital approach to health care surveillance which will complement routine reporting and enhance outbreak detection across the continent. In conclusion, these studies showcase a clear progression in Sub-Saharan Africa: coming from the recognition of potential of digital tools, to assessing the performance and limitations of existing surveillance systems, to the practicability of search-query data use in both infectious and non-communicable disease contexts.

1.4.3 LOCAL PERSPECTIVE

Although the integration of digital health in surveillance systems has evolved in other regions, Zambia exhibits a considerable gap in this context. It is evident that there is not much literature under this field; however, research has suggested that individuals, particularly mothers, turn to the internet for health-related information when access to healthcare is limited (Mulauzi *et al.*, 2020).

At the same time, Zambia Information and Communications Technology Authority (ZICTA) disclosed Zambia's overall internet subscriptions growth by 7.0% in 2024 – from 12.6 million in 2023 to 13.5 million in 2024, raising the internet-penetration rate from 64.2% to 67.3% of the population. Similarly, the Presidential Delivery Unit noted that as of 2025, 50% of all Zambians have basic digital literacy. These accounts point to a research gap and suggest the expanding potential for digital health surveillance systems in the region.

1.5 GENERAL OBJECTIVE

To assess the correlation between Google RSV and EID burden in Zambia from 2019 and 2024.

1.5.1 SPECIFIC OBJECTIVES

1. To analyze trends EID search queries in Zambia from 2019 to 2024.
2. To examine trends in EID burden in Zambia over the same period.
3. To examine the influence of contextual factors such as geographic variations and internet penetration, population size, and major health events on EID search behavior.
4. To evaluate Google RSV's potential as a supplementary surveillance tool through its correlation with EID burden in Zambia.

1.6 RESEARCH QUESTIONS

1. What are the trends in EID Google search queries in Zambia from 2019 to 2024?
2. What are the trends in reported EID cases over the same period?
3. How do contextual factors such as internet penetration, population size, and major health events influence EID search behavior?
4. How does Google RSV correlate with EID burden, and can it be used as a supplementary surveillance tool in Zambia from 2019 to 2024?

1.7 MEASUREMENTS

The measurements will focus on using variable keys to track EIDs Google searches as the main outcome, looking at trends over time and across regions in Zambia, and weighed against actual EID prevalence. Factors like population size, internet access, and health campaigns are contextual and will help interpret the variations in search activity.

VARIABLE	DEFINITION	DATA SOURCE	TYPE
Google Trends RSV	Relative Search Volume (RSV) for selected EID over time.	Google Trends	Independent (Predictor)
EID burden	Number of reported EID cases within the same period.	HMIS/DHIS2	Dependent (Outcome)
Time	Month and year of observation (2019-2024)	Google Trends / HMIS	Temporal dimension
Geography	Provincial distribution of search data and reported cases	Google Trends / HMIS	Spatial dimension
Population size	Number of people in region of interest	Zambia CSO 2023 projections	Covariate
Internet Penetration	Number of people with internet access per region of interest.	ZICTA annual report from 2019 - 2024	Covariate
Major health events/campaigns	Public health campaigns or events influencing search behavior	Media reports / MOH	Covariate

Table 1 Measurements

1.8 THEORETICAL FRAMEWORK

This study is guided by two primary theories: the theory of Planned Behaviour (TPB) and the Digital Phenotyping framework.

TPB human behaviour argues that human behaviour is a consequence of three key factors: attitude toward the behaviour, subjective norms, and perceived behavioural control. In relation to health information-seeking, this theory suggests that individuals are likely to search for health-information if they believe it is beneficial (attitude), recognize that peers do the same (subjective norms), and appreciate how easy it is to access and use digital resources (perceived behavioural control).

The digital Phenotyping framework suggests that online digital traces, such as search queries, can reveal insights into population-level health behaviours and awareness of health risks. Analysis of these digital footprints can serve as a proxy for monitoring public health trends.

To elaborate, the combination of these two theories, this study is likely to address both behavioural incentives behind online searches and the assessment of these queries as public health indicators for EID incidence. Conclusively, this theoretical approach provides a firm foundation for assessing the feasibility of using search query data to supplement traditional surveillance in Zambia.

1.9 CONCEPTUAL FRAMEWORK

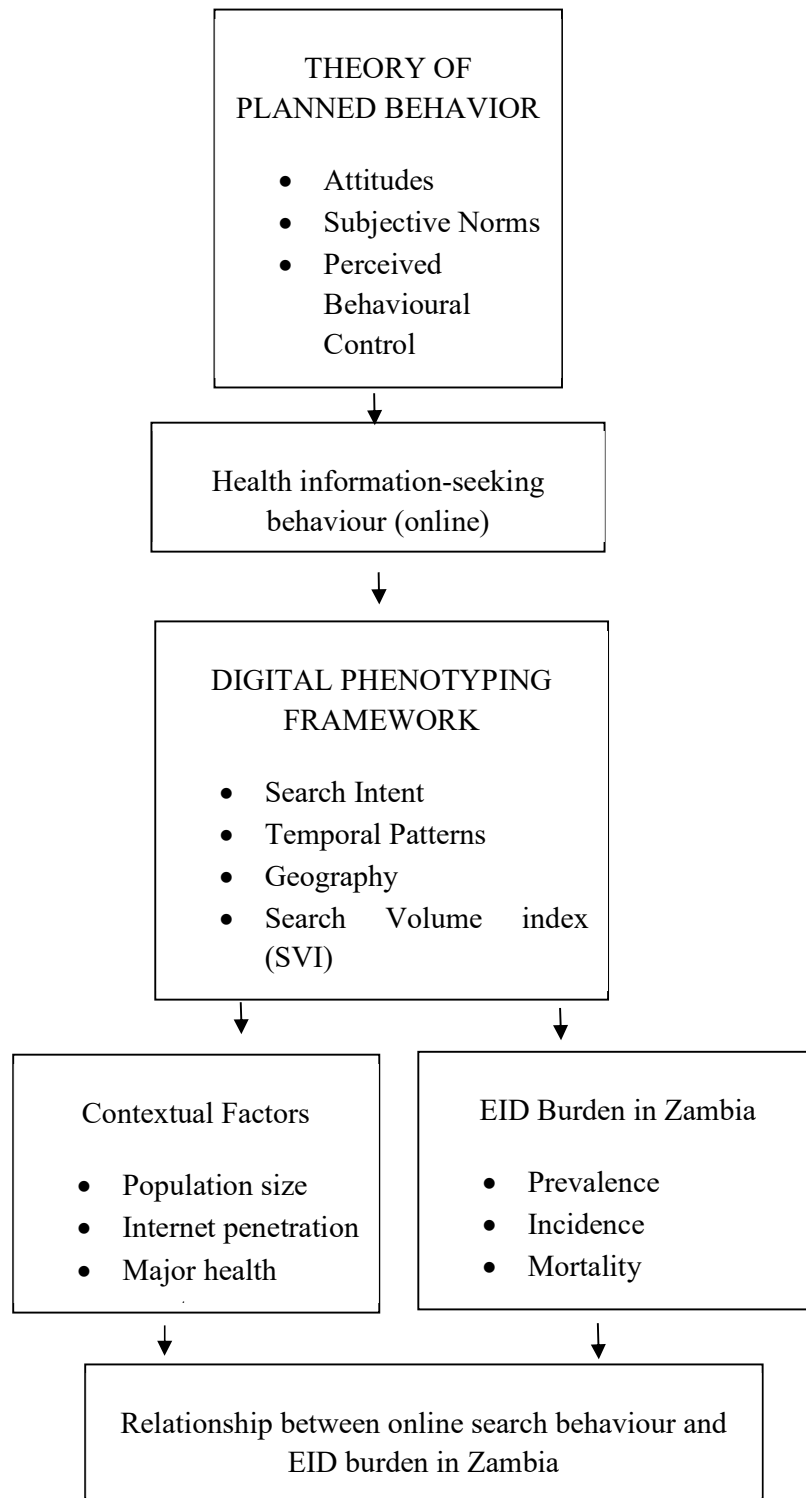


Figure 1 Conceptual Framework

CHAPTER TWO METHODOLOGY

2.1 STUDY SITE

This study will be conducted using aggregated GST data for Zambia and official EID burden data for Zambia covering the period 2019-2024.

2.2 STUDY POPULATION

Data will be drawn from two main datasets. The first dataset will be Google Trends data, which provides RSV on a 0–100 scale for selected EID-related search terms in Zambia. The second dataset will be HMIS/DHIS2 EID records, which will serve as the official measure EID burden for the same period. A third contextual dataset may be added for interpretation and adjustment, including ZICTA internet penetration and CSO population estimates.

2.3 STUDY DESIGN

This will be a quantitative, correlational ecological study using secondary time-series data. The design will assess whether monthly Google RSV for EID-related terms is associated with monthly EID burden in Zambia.

2.4 INCLUSION/EXCLUSION CRITERIA

2.4.1 Inclusion Criteria

Dataset 1 – Google Relative Search Volume (RSV) Data

Source: Google Trends via Pytrends, with the geo marker set to = ‘ZM’

1. Search terms must directly name a primary disease of interest:
 - Cholera = [‘Cholera’, ‘Cholera symptoms’, ‘Cholera Zambia’]
 - Typhoid = [‘Typhoid’, ‘Typhoid fever’, ‘typhoid symptoms’]
 - Ebola = [‘ebola’, ‘ebola outbreak’, ‘ebola symptoms’]
2. Search terms must directly name a secondary disease of interest:
 - Mpox = [‘mpox’, ‘monkeypox’, ‘monkeypox symptoms’]
 - Measles = [‘measles’, ‘measles rash’, ‘measles Zambia’]
3. Data must fall within the study period (2019-2024)

Dataset 2 – Emerging Infectious Disease Case Reporting Data

Source: Zambia Health Management Information System (HMIS) and District Health Information System 2 (DHIS2) data for primary and secondary diseases included in Google RSV data.

2.4.2 Exclusion

Incomplete or missing time-series data

Non-EIDs-related search terms

Data not aligned to the study period

2.5 DATA COLLECTION

Data Sources

The study will make use of secondary data from two main sources:

1. Google Relative Search Volume Data (RSV):

EIDs-related search queries in Zambia between 2019 and 2024 will be extracted from Google Trends.

- Search terms will be identified based on 2.4.1 Inclusion Criteria.
- Data will be collected on a monthly and regional basis to allow temporal and geographic analysis.
 - i. Location: Zambia
 - ii. Time range: January 2019 to December 2024
 - iii. Category: Health
 - iv. Search type: Web search
 - v. Time resolution: Monthly

To improve reliability, multiple related search terms will be analyzed and, where appropriate, combined into a single composite RSV index by averaging normalized values across selected terms.

2. HMIS/DHIS2 Data

- Official Health Management Information System (HMIS) and District Health Information System 2 (DHIS2) data on included EID-reported cases in Zambia will be obtained for the same period.
- The data will be gathered at the national and, when available, provincial levels and will be combined into monthly time-series to match Google Trends data.

3. Contextual Data

Additional contextual variables will be collected to support the interpretation of results.

These include:

- Internet penetration data from ZICTA reports
- Population estimates from the Central Statistical Office (CSO)
- Major health events or campaigns obtained from the MOH reports and relevant media sources

These variables will be used to provide context and explain variations in search behavior over time and across regions.

2.6 DATA ANALYSIS

Data analysis will be conducted using Python. The following libraries will be used:

- Pandas: for data cleaning, merging, and manipulation.
- Matplotlib & Seaborn: for plotting trends and heatmaps.
- Statsmodels & scipy: for correlation and regression analysis.
- Geopandas: for mapping regional data.

Data Preparation

- **Objective:** To prepare datasets for analysis
- **Method:** Google RSV and HMIS/DHIS2 EID case data will be imported into Python and cleaned using Pandas. Given that GST data is provided as normalized values on 0-100 scale, it will be used as relative indicators of search interest. To ensure comparability between variables, each data set will be assigned ranks for the sake of Spearman's rho. Missing values will be handled, and both datasets will be aggregated into a monthly time-series format (2019–2024). The datasets will then be merged based on time (month and year) to allow direct comparison.

Descriptive and Temporal Trend Analysis

- **Objective:** To examine patterns and trends over time
- **Method:** Line graphs will be plotted to visualise monthly trends in Google search interest and reported EID cases. Visual inspection will be used to identify similarities, peaks, and fluctuations in both datasets over time.

Geographical Analysis

- **Objective:** to assess regional differences in search behavior.
- **Method:** Google Trends data will be mapped at the provincial level using GeoPandas.

Correlational Analysis

- **Objective:** assess the strength and direction of associations between EIDs-related search queries and EID cases.

- **Method:** Both Pearson's correlation coefficient and Spearman's rank will be calculated to measure the strength and direction of the relationship between Google Trends Relative Search Volume (RSV) and reported EID cases. Pearson's correlation will be used to assess linear associations between continuous variables, while Spearman's rho will be used as a non-parametric measure that is robust to non-normal distributions and appropriate for ranked or normalized data such as RSV.

The correlation coefficient (r) will be interpreted as follows:

- **r close to +1:** strong positive correlation
- **r close to 0:** no correlation
- **r close to -1:** negative correlation

Statistical significance (p -value < 0.05) will be used to determine whether the observed relationship is meaningful.

Lagged Correlational Analysis

- **Objective:** To assess whether Google searches occur before reported EID cases.
- **Method:** A lagged correlation analysis will be conducted by shifting the Google Trends data forward by 1–3 months and recalculating the correlation with EID cases.

Regression Analysis

- **Objective:** To assess whether Google Search Interest can predict EID cases.
- **Method:** A simple linear regression model will be used, where:
 - Independent variable: Google Trends RSV
 - Dependent variable: reported EID cases

2.7 ETHICAL CONSIDERATIONS

Although involving secondary, aggregated data rather than direct human subjects, this study will follow ethical standards in public health research. Nonetheless, since this data does not contain any personal or identifiable information and there is no direct involvement of human participants, issues such as informed consent are not applicable. The use of such data ensures that individual privacy and confidentiality are protected throughout the study.

However, ethical responsibility in this research goes beyond data privacy. Care will be taken to ensure that the findings are interpreted correctly and not exaggerated, especially since online search behavior can be influenced by factors such as media coverage, public awareness campaigns, or changes in internet access. The study also recognizes that Google Trends data may not fully represent populations with limited internet access, which is an important consideration in the Zambian context. All data sources will be properly cited and securely stored with restricted access by the researcher, and the study will follow institutional research guidelines, including obtaining ethical approval from CBU-BREC.

2.8 STUDY LIMITATIONS

Despite the strengths of this study, several methodological and data-related limitations should be acknowledged, as they may influence the interpretation and generalizability of the findings. Google Trends provides relative search volume rather than absolute search counts, reflecting patterns in search interest over time rather than the actual number of people searching for EIDs-related information. This limits the ability to directly quantify public interest or compare search intensity across different contexts.

Furthermore, facility-level reporting in Zambia is inconsistent, and EIDs may be under-reported, especially in rural areas. This might mean that the dependent variable does not fully reflect the true burden of disease.

Finally, this study emphasizes correlation instead of causation, and associations observed at the population level may not be true at the individual level.

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APPENDICES

APPENDIX I: GLOSSARY

Cholera	An acute diarrheal infection caused by ingestion of food or water contaminated with the bacterium <i>Vibrio cholerae</i> . Classified as a primary emerging infectious disease of interest in this study due to recurrent outbreaks in Zambia.
Composite RSV Index	A combined measure created by averaging the relative search volumes of multiple related search terms for a single disease, used to improve the reliability of Google Trends data in analysis.
Contextual Factors	External variables — including internet access, population size, and major public health events — that may independently influence observed search behaviour and must be accounted for when interpreting RSV data.
Correlation	A statistical measure that describes the strength and direction of a relationship between two variables. A positive correlation means both variables increase together; a negative correlation means one increases as the other decreases.
COVID-19	A respiratory illness caused by the SARS-CoV-2 coronavirus, declared a pandemic by WHO in March 2020. Classified as a primary emerging infectious disease of interest in this study.
DHIS2 (District Health Information Software 2)	An open-source, web-based health information management platform used by the Zambia Ministry of Health to collect, manage, and analyse routine health facility data, including disease surveillance reports.

Digital Epidemiology	The use of digital data sources — such as internet search queries, social media, and mobile data — to monitor and analyse disease patterns at the population level.
Digital Phenotyping	A framework proposing that digital traces left by individuals online — such as search queries — can serve as measurable proxies for population-level health behaviours, awareness, and concerns.
Ebola	A severe and often fatal hemorrhagic fever caused by the Ebola virus. Included in this study due to its status as a high-consequence pathogen with documented spillover risk from neighbouring Democratic Republic of Congo (DRC).
Ecological Study	A study design that examines data at the population or group level rather than the individual level. In this study, both RSV and disease case data are aggregate national or provincial figures, not individual records.
Emerging Infectious Disease (EID)	An infectious disease that has either appeared and affected a population for the first time, or has previously existed but is rapidly increasing in incidence, geographic range, or both (WHO, 2016).
Google Trends	A freely accessible online tool developed by Google that analyses the relative popularity of search queries across different geographic regions and time periods, outputting data as a normalised Relative Search Volume (RSV) index on a scale of 0 to 100.
HMIS (Health Management Information System)	A system for collecting, storing, managing, and transmitting health data. In Zambia, the HMIS operates through DHIS2 and serves as the primary repository for routine facility-based disease reporting.

Infodemiology	The study of the distribution and determinants of health-related information on the internet and its implications for public health practice and policy.
Internet Penetration	The proportion of a population with active access to the internet. In this study, internet penetration data from ZICTA is used as a covariate to contextualise variations in RSV across time and regions.
Lagged Correlation	A statistical method used to determine whether changes in one variable consistently precede changes in another over time. In this study, it tests whether RSV spikes occur before increases in reported EID cases, assessing the early warning potential of Google Trends.
Measles	A highly contagious viral infection caused by the measles virus. Classified as a secondary disease of interest in this study due to recurrent outbreak cycles in Zambia linked to immunisation gaps.
Mpox (Monkeypox)	A viral zoonotic disease caused by the monkeypox virus. Classified as a secondary disease of interest following its declaration as a Public Health Emergency of International Concern (PHEIC) by WHO in 2022 and renewed designation in 2024.
Pearson's Correlation Coefficient (r)	A parametric statistical measure of the linear relationship between two continuous variables. Assumes normal distribution of data. Values range from -1 to +1, where +1 indicates a perfect positive relationship.
Public Health Surveillance	The continuous, systematic collection, analysis, and interpretation of health-related data, used for planning, implementing, and evaluating public health interventions.

Relative Search Volume (RSV)	A normalised measure produced by Google Trends, ranging from 0 to 100, representing the popularity of a search term relative to its own peak within a specified time period and geographic region. It is a relative indicator, not an absolute count of searches.
Spearman's Rank Correlation Coefficient (rho)	A non-parametric statistical measure of the strength and direction of association between two ranked variables. Unlike Pearson's r , it does not assume normal distribution and is appropriate for RSV data, which is ordinal in nature.
Theory of Planned Behaviour (TPB)	A psychological theory proposing that an individual's behaviour is determined by their attitude toward the behaviour, subjective norms (perceived social expectations), and perceived behavioural control (belief in ability to perform the behaviour).
Time-Series Data	Data collected at successive, equally-spaced time intervals. In this study, monthly observations from January 2019 to December 2024 are used to analyse trends and temporal relationships between RSV and EID case data.

APPENDIX II: BUDGET

Item	Details	Estimated Cost (ZMW)
Data Access	Access to DHIS/HMIS and Google Trends data	K 0.00
Software & Tools	Python and libraries	K 0.00
Ethical Clearance Fees	IRB processing	K 500.00
Miscellaneous Expenses	Transport, Communication, utilities.	K 6,000.00
Computational Resources	Cloud storage, server costs	K 6,250.00
Total		K 12,750.00

Table 2 Budget

APPENDIX III: GANTT CHART

Activity / Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Proposal revision & corrections	■	■						
Search term operationalisation	■							
Ethics application (CBU-BREC)	■	■						
MOH data access letter		■						
★ Ethics submitted		★						
Google Trends data extraction			■	■				
DHIS2/HMIS data request			■	■				
Contextual data (ZICTA, CSO)			■	■				
★ All datasets collected				★				

Activity / Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Data cleaning & merging (Python)				■	■			
Descriptive & trend analysis					■			
Pearson / Spearman correlation					■	■		
Lagged correlation (1–3 months)						■		
Regression modelling						■		
Geographic (provincial) mapping						■		
★ Analysis complete						★		
Results write-up & tables/figures						■	■	
Discussion & conclusion drafting							■	

Activity / Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Supervisor review & feedback							■	■
Revisions & final formatting								■
★ Final submission								★

Figure 2 Grantt Chart

APPENDIX IV: INFORMATION SHEET

Title: Evaluating Online Health-Seeking Behaviors as Digital Proxies for Emerging Infectious Disease Burden in Zambia: A Google Trends Time-Series Analysis (2019–2024)

Purpose of Study: This study aims to analyze the correlation between Google RSV data and EID burden in Zambia. It focuses on the integration of search query data and public digital phenotypes as an indicator in public health surveillance.

Participant Involvement: This study utilizes secondary aggregated data and does not involve direct participant interaction.

Benefits of the Study: The findings of this study can lead to improved public health strategies and more effective interventions.

Risks: As this study involves aggregated data, risks are minimal. And measures are in place to ensure the anonymized data is secure.

Confidentiality: Collected data will be anonymized, stored securely, and only accessed by authorized personnel. The study complies with the General Data Protection Regulation (GDPR) and the Data Protection Act of Zambia.

Voluntary Nature of Participation: This study utilizes secondary aggregated data and does not involve direct participant interaction.

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